

Lab 04 - Algorithm and Algorithm Efficiency

There are 2 questions in this lab. You will make a copy of this document for your work. Submit your work to the Lab 04 Assignment in D2L. Solutions will be posted the day after the lab is due and only for students who submit work. Check your answers - learning to self-assess is super important and will help you become a better computer scientist!

1. Consider the following lists of partially sorted numbers using Insertion Sort. The double bars represent the sort marker. **For each show the sorting process step by step until the next number is sorted, like in the supplemental video.**

1. [1 3 4 8 9 || 5 2]
2. [1 2 4 7 || 6 10]

Hint: Check the insertion sort algorithm in L07 and L08.

2. Consider the following lists of partially sorted numbers using Selection Sort. The double bars represent the sort marker. **For each show the sorting process step by step until the next number is sorted, like in the supplemental video.**

1. [1 3 4 6 7 || 9 8]
2. [2 3 4 || 9 8 7 6 5]

Hint: Check the selection sort algorithm in L05 and L06.